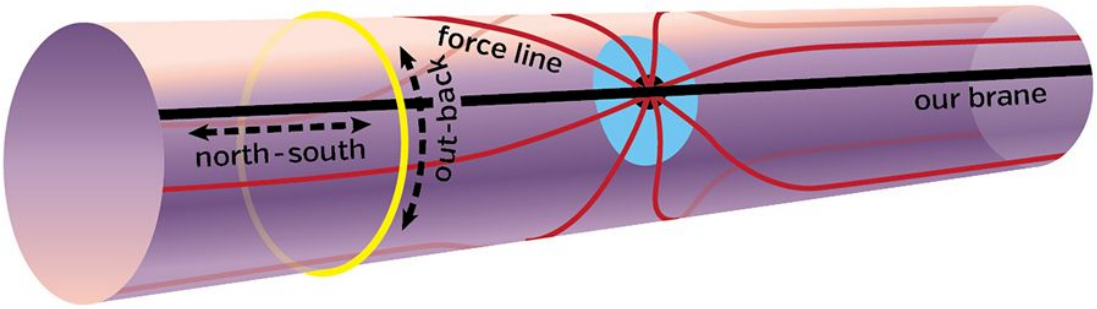


Fig. 23.3. If the out-back dimension (*yellow*) is curled up, then outside the blue circle a particle's gravitational force lines (*red*) are parallel to our brane.

Figure 23.3 depicts this for the gravity of a tiny particle that resides at the center of the blue disk. In this picture, two space dimensions are suppressed, so we see only one of our brane's dimensions (call it north-south) along with the bulk's out-back dimension. Near the particle, inside the blue disk, the force lines spread in the out-back dimension as well as north-south, so (with the missing dimensions restored) gravity's strength obeys an inverse cube law. However, outside the blue disk the curl-up makes the force lines lie parallel to our brane. They spread no further into out-back, and Newton's inverse square law is restored.

Physicists who struggle to understand quantum gravity think this is the fate of all the extra dimensions except possibly one or two: they are curled up on microscopic scales, preventing gravity from spreading too fast. In *Interstellar*, Christopher Nolan ignores these curled-up dimensions and focuses on just one bulk dimension that's not curled up. This becomes his out-back, fifth dimension.

Why should out-back not be curled up? For Chris the answer is simple: A curled-up bulk has very little volume—nowhere near enough volume to be an arena for interesting science fiction. For Cooper to travel into the bulk riding in the tesseract, as he does in the movie, the tesseract needs far more volume than a curled-up dimension would provide.